

Odd cycles in coprime graphs: an answer to a question of Erdős and Sárközy

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Abstract: We prove that for every sufficiently large n , every set $A \subseteq \{1, \dots, n\}$ with more than $\lfloor n/2 \rfloor + \lfloor n/3 \rfloor - \lfloor n/6 \rfloor$ elements has a coprime graph containing cycles of all odd lengths from 3 to $n/3 + 1$. This answers a question of Erdős and Sárközy (Erdős Problem 883). The proof is formally verified in Lean 4.

Keywords: coprime graph; odd cycles; formal verification

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1. Introduction

For a finite set A of positive integers, the *coprime graph* $G(A)$ has vertex set A , two distinct elements being adjacent when they are coprime. A cycle of length $L \geq 3$ in $G(A)$ is a sequence of L distinct elements $v_1, \dots, v_L$ of A such that v_i is coprime to v_{i+1} for $i < L$ and v_L is coprime to v_1 . Write $[n] = \{1, \dots, n\}$ and

$$T(n) = \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor$$

for the number of elements of $[n]$ divisible by 2 or by 3. The set $M(n)$ of these elements spans no triangle, since among any three of them two are even or two are multiples of 3.¹

¹The extremal role of the multiples of the first primes goes back to Erdős [3]. Let $f(n, k)$ be the number of integers in $[n]$ divisible by at least one of the first k primes, so that $f(n, 2) = T(n)$. Erdős asked whether every $A \subseteq [n]$ with $|A| > f(n, k)$

Erdős and Sárközy [4] proved that one additional element forces much more than a triangle: there is a constant $c > 0$ such that, for every large n and every $A \subseteq [n]$ with $|A| > T(n)$, the graph $G(A)$ contains a cycle of length $2l + 1$ for every positive integer $l \leq cn$. They asked whether $c = 1/6$ is admissible. Since $l \leq n/6$ means $2l + 1 \leq n/3 + 1$, this is Question 1 of Erdős Problem 883 [2].² We answer the question.

Theorem 1.1. *There is an integer n_0 such that for every $n \geq n_0$ and every $A \subseteq [n]$ with $|A| > T(n)$, the coprime graph $G(A)$ contains a cycle of length L for every odd L with $3 \leq L \leq n/3 + 1$.*

The size condition is sharp, and so is the range of lengths when $6 \mid n$. The set $M(n)$ shows that $|A| > T(n)$ cannot be weakened. Two even numbers are never adjacent, so the even vertices of a cycle of odd length L are pairwise non-consecutive and there are at most $(L - 1)/2$ of them; such a cycle contains at least $(L + 1)/2$ odd numbers. The set $M(n) \cup \{1\}$ has $T(n) + 1$ elements but only $o(n) + 1$ odd ones, where

$$o(n) = \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor$$

is the number of odd multiples of 3 in $[n]$, so its odd cycles have length at most $2o(n) + 1$, which equals $n/3 + 1$ when $6 \mid n$.

The approach. Let d be the number of integers $w \leq n$ with $w \equiv 2, 4 \pmod{6}$ that are missing from A . When $70d \leq n$, the even numbers prime to 3 that lie in A form a large reservoir of connecting vertices. Every odd length from 5 to $2o(n) + 1$ is then obtained by joining short paths through odd vertices with the help of Jackson’s theorem on cycles in bipartite graphs (Theorem 6.1). When $70d > n$ this reservoir is too small. Depending on how many multiples of 2 or 3 are missing from A , we either use all multiples of 2 or 3 that lie in A as connectors between units of A (Proposition 7.2), or keep the even numbers prime to 3 and add one long path that alternates between units of A and multiples of 6 in A (Proposition 7.3). Both constructions need many elements z of A with a large value of

$$\rho'(z) = \prod_{p \mid z, p \geq 5} \left(1 - \frac{1}{p}\right),$$

and the exceptions in each residue class modulo 6 are bounded by sixteen explicit density certificates (Theorem 4.4). A finite cover of the parameter space by boxes (Section 8) shows that one of the two constructions applies in every case. Figure 1 shows the structure of the proof.

Formal verification. Theorem 1.1 is proved in Lean 4 with the Mathlib library [6, 7]. The formal statement is quoted in Section 9. Its proof uses only the three standard axioms of Lean and Mathlib, and every numerical inequality, the sixteen certificates included, is checked by the Lean kernel.³ The text follows the formal proof step by step.

2. Notation and counting in residue classes

Throughout, p denotes a prime. For an integer $m \geq 1$ let $\omega'(m)$ be the number of primes $p \geq 5$ dividing m , and let $\rho'(m) = \prod_{p \mid m, p \geq 5} (1 - 1/p)$, with $\rho'(m) = 1$ when $\omega'(m) = 0$.

contains $k + 1$ pairwise coprime integers, that is, a complete subgraph on $k + 1$ vertices in $G(A)$. For $k = 2$ the answer is yes (this is Theorem 5.1 below), while Ahlswede and Khachatrian [1] showed that the answer is negative for some larger k . Erdős and Sárközy [4] replaced complete subgraphs by odd cycles.

²Erdős and Sárközy state their theorem, and the question, for $n \geq n_0$. The problem list [2] records the question without the words “for large n ”. Theorem 1.1 answers the question in the form posed in [4], for all n from some point on; it says nothing about small n .

³Lean is an interactive theorem prover based on dependent type theory. A proof is accepted only after the kernel, a small program that implements the typing rules, has checked every step. The command `#print axioms` lists the axioms a theorem depends on; for Theorem 1.1 the list consists of propositional extensionality, the axiom of choice and the soundness of quotients, which every development based on Mathlib uses. The numerical certificates are evaluated by the kernel’s own reduction (`decide +kernel`) and do not rely on compiled code.

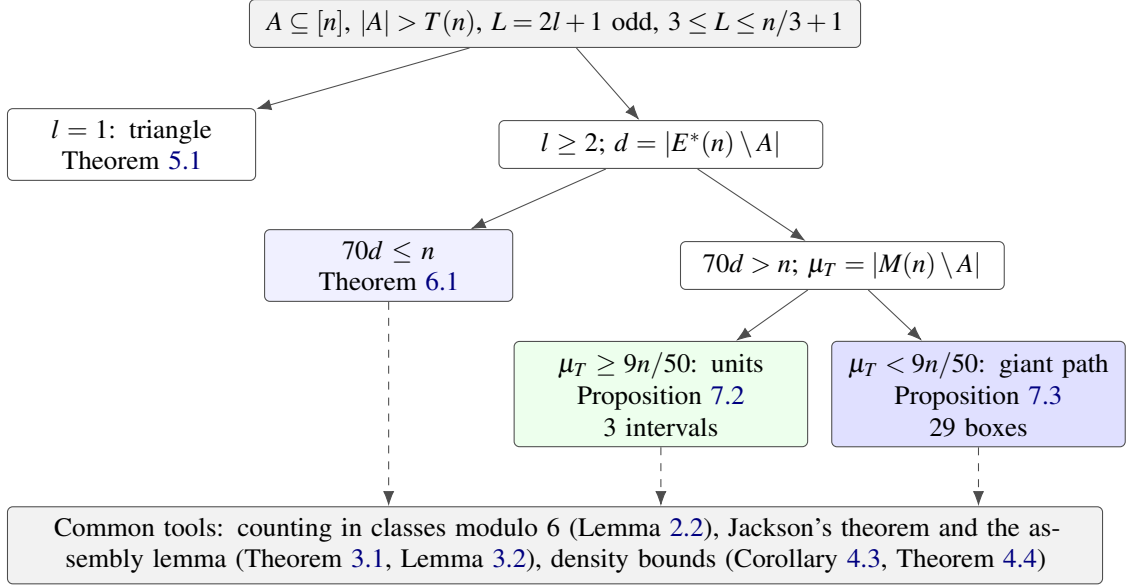


Figure 1: Structure of the proof of Theorem 1.1.

2.1. Residue classes modulo 6

We split $[n]$ into the classes

$$U(n) = \{z : z \equiv \pm 1\}, \quad O_3(n) = \{z : z \equiv 3\}, \quad E^*(n) = \{z : z \equiv 2, 4\}, \quad M_6(n) = \{z : z \equiv 0\},$$

congruences being modulo 6. Thus $M(n) = O_3(n) \cup E^*(n) \cup M_6(n) = [n] \setminus U(n)$ and $|M(n)| = T(n)$, while $|E^*(n)| = \lfloor n/2 \rfloor - \lfloor n/6 \rfloor$ and $|M_6(n)| = \lfloor n/6 \rfloor$. Elements of $U(n)$ are called *units*. For $A \subseteq [n]$ we write

$$d = |E^*(n) \setminus A|, \quad \mu_6 = |M_6(n) \setminus A|, \quad \mu_T = |M(n) \setminus A|, \quad u = |A \cap U(n)|.$$

Whether two integers can be coprime is decided by their residues modulo 6: they cannot be coprime when both are even or both are multiples of 3. Formally, let $c : [n] \rightarrow \mathbb{Z}/6\mathbb{Z}$ be reduction modulo 6, let $\gamma(x, y) = 1$ if $\gcd(x, y, 6) = 1$ and $\gamma(x, y) = 0$ otherwise, and define κ on $(\mathbb{Z}/6\mathbb{Z})^2$ in the same way. Since $\gcd(x, y, 6)$ depends only on x and y modulo 6, the triangle

$$\begin{array}{ccc} [n] \times [n] & \xrightarrow{c \times c} & (\mathbb{Z}/6\mathbb{Z})^2 \\ & \searrow \gamma & \downarrow \kappa \\ & & \{0, 1\} \end{array} \quad (1)$$

commutes, and $\gcd(x, y) = 1$ implies $\gamma(x, y) = 1$. Figure 2 shows the pairs of classes on which $\kappa = 1$. The classes inside $M(n)$ are joined only by the edge between O_3 and E^* , which is another way to see that $G(M(n))$ is bipartite and has no triangle.

Lemma 2.1. *Let $A \subseteq [n]$ with $|A| > T(n)$, and let D be a set of even numbers in $[n]$ disjoint from A . Then A has at least $o(n) + 1 + |D|$ odd elements.*

Proof. The even elements of A lie among the $\lfloor n/2 \rfloor - |D|$ even numbers of $[n]$ outside D . Hence A has at least $T(n) + 1 - \lfloor n/2 \rfloor + |D| = o(n) + 1 + |D|$ odd elements. $\square$

2.2. Counting coprime elements of a class

A set $R \subseteq \mathbb{Z}/6\mathbb{Z}$ is *symmetric* if $R = -R$. For a finite set P of positive integers and an integer $x \geq 0$ put

$$C_R(x; P) = \#\{w \in [x] : w \bmod 6 \in R, p \nmid w \text{ for all } p \in P\}.$$

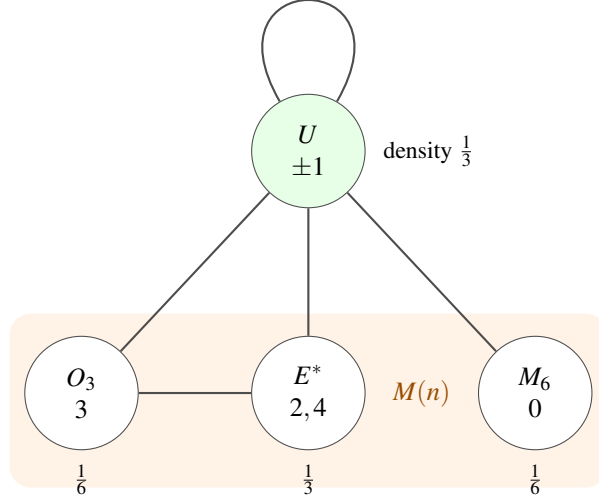


Figure 2: The four residue classes modulo 6 and their densities. An edge, or the loop at U , marks the pairs of classes that contain coprime pairs. The shaded region is $M(n)$.

Lemma 2.2. *Let R be symmetric and let $c_R \geq 0$ satisfy $|C_R(x; \emptyset) - x|R|/6| \leq c_R$ for all integers $x \geq 0$. Let P be a finite set of pairwise coprime integers, each $\equiv \pm 1 \pmod{6}$. Then for every $n \geq 0$,*

$$\left| C_R(n; P) - \frac{n}{6} |R| \prod_{p \in P} \left(1 - \frac{1}{p}\right) \right| \leq \left(c_R + \frac{|R|}{6} \right) 2^{|P|} - \frac{|R|}{6}.$$

One may take $c_R = 5|R|/6$ for every R , and $c_R = 2/3$ for $R = \{2, 4\}$.

Proof. The class of r meets $[x]$ in $\lfloor (x + 6 - r')/6 \rfloor$ elements, where $r' \in \{1, \dots, 6\}$ represents r ; this differs from $x/6$ by at most $5/6$, which gives $c_R = 5|R|/6$. For $R = \{2, 4\}$ we have $C_R(x; \emptyset) = \lfloor x/2 \rfloor - \lfloor x/6 \rfloor$, which differs from $x/3$ by at most $2/3$.

We argue by induction on $|P|$; the case $P = \emptyset$ is the hypothesis. Let $q \notin P$ be coprime to every element of P , with $q \equiv \pm 1 \pmod{6}$. Multiplication by q maps $\lfloor [n/q] \rfloor$ bijectively onto the multiples of q in $[n]$, and since $q \equiv \pm 1$ and $R = -R$, the square

$$\begin{array}{ccc} \lfloor [n/q] \rfloor & \xrightarrow{w \mapsto qw} & [n] \\ \text{mod } 6 \downarrow & & \downarrow \text{mod } 6 \\ R & \xrightarrow{r \mapsto qr} & R \end{array} \quad (2)$$

is defined on the elements whose residues lie in R and commutes. For $p \in P$ we have $p \mid qw$ if and only if $p \mid w$, because $\gcd(p, q) = 1$. Hence the elements counted by $C_R(n; P)$ that are multiples of q are exactly the numbers qw with w counted by $C_R(\lfloor n/q \rfloor; P)$, and

$$C_R(n; P \cup \{q\}) = C_R(n; P) - C_R(\lfloor n/q \rfloor; P).$$

Write $\Pi = \prod_{p \in P} (1 - 1/p) \in (0, 1]$ and $E_P(x) = |C_R(x; P) - x|R|\Pi/6|$. The main term for $P \cup \{q\}$ equals $n|R|\Pi/6 - (n/q)|R|\Pi/6$, and $|(n/q) - \lfloor n/q \rfloor| < 1$. Therefore

$$E_{P \cup \{q\}}(n) \leq E_P(n) + E_P(\lfloor n/q \rfloor) + \frac{|R|}{6},$$

so that $E_{P \cup \{q\}}(n) + |R|/6 \leq 2 \max_x (E_P(x) + |R|/6) \leq 2^{|P|+1} (c_R + |R|/6)$ by the induction hypothesis. $\square$

Write $\mathcal{P}(m)$ for the set of primes $p \geq 5$ dividing m , and $N^*(n, m) = \{w \in E^*(n) : \gcd(w, m) = 1\}$.

Corollary 2.3. Let $m \geq 1$ and let R be symmetric. Suppose that every element of R is odd if $2 \mid m$, and prime to 3 if $3 \mid m$. Then

$$\left| \#\{w \in [n] : w \bmod 6 \in R, \gcd(w, m) = 1\} - \frac{n}{6} |R| \rho'(m) \right| \leq |R| \left(2^{\omega'(m)} - \frac{1}{6} \right).$$

For odd m , moreover, $\left| |N^*(n, m)| - \frac{n}{3} \rho'(m) \right| \leq 2^{\omega'(m)} - \frac{1}{3}$.

Proof. Under the hypothesis, an integer w with $w \bmod 6 \in R$ is coprime to m exactly when no prime of $\mathcal{P}(m)$ divides it, so the count equals $C_R(n; \mathcal{P}(m))$. Every prime $p \geq 5$ is $\equiv \pm 1 \pmod{6}$, and $\prod_{p \in \mathcal{P}(m)} (1 - 1/p) = \rho'(m)$. Lemma 2.2 with $c_R = 5|R|/6$, and with $c_R = 2/3$ for $R = \{2, 4\}$, gives both bounds. $\square$

2.3. The density ρ'

Lemma 2.4. Let $a, b, m, z \geq 1$.

- (a) $\rho'(m) \geq 1/(\omega'(m) + 1)$ and $(\omega'(m) + 1)! \leq m$.
- (b) If $z \mid m$, then $\rho'(m) \leq \rho'(z)$.
- (c) $\rho'(\text{lcm}(a, b)) \geq \rho'(a)\rho'(b)$.
- (d) $N^*(n, \text{lcm}(a, b)) = N^*(n, a) \cap N^*(n, b)$, that is, the square of inclusions

$$\begin{array}{ccc} N^*(n, \text{lcm}(a, b)) & \hookrightarrow & N^*(n, a) \\ \downarrow & & \downarrow \\ N^*(n, b) & \hookrightarrow & E^*(n) \end{array} \quad (3)$$

is a pullback.

Proof. (a) Let $p_1 < \dots < p_K$ be the primes of $\mathcal{P}(m)$, $K = \omega'(m)$. Then $p_i \geq i + 1$, so $\rho'(m) \geq \prod_{i \leq K} (1 - 1/(i + 1)) = 1/(K + 1)$ and $m \geq \prod_i p_i \geq (K + 1)!$. (b) holds because $\mathcal{P}(z) \subseteq \mathcal{P}(m)$ and every factor is at most 1. (c) Since $\mathcal{P}(\text{lcm}(a, b)) = \mathcal{P}(a) \cup \mathcal{P}(b)$, the product over the union equals the product over $\mathcal{P}(a)$ times the product over $\mathcal{P}(b)$ divided by the product over $\mathcal{P}(a) \cap \mathcal{P}(b)$, which is at most 1. (d) An integer is coprime to $\text{lcm}(a, b)$ if and only if it is coprime to a and to b . $\square$

Lemma 2.5. Let $N_0 = 1000 \cdot 23 \cdot 2^{22}$ and $N_1 = 10^6 \cdot 30 \cdot 2^{29}$, and let $1 \leq m \leq n^2$. If $n \geq N_0$, then $1000(\omega'(m) + 1)2^{\omega'(m)} \leq n$. If $n \geq N_1$, then $10^6(\omega'(m) + 1)2^{\omega'(m)} \leq n$.

Proof. Let $K = \omega'(m)$ and $h(K) = 1000(K + 1)2^K$. If $K \leq 21$, then $h(K) \leq N_0$. For $K \geq 22$ we show $h(K)^2 \leq (K + 1)!$; together with Lemma 2.4(a) this gives $h(K)^2 \leq m \leq n^2$. For $K = 22$ the inequality reads $(9.65 \dots \cdot 10^{10})^2 \leq 23! = 2.58 \dots \cdot 10^{22}$. Passing from K to $K + 1$ multiplies the left side by $4(K + 2)^2/(K + 1)^2$ and the right side by $K + 2$, and $4(K + 2) \leq (K + 1)^2$ for $K \geq 4$. The second statement is proved in the same way with 10^6 in place of 1000, the cut at $K = 29$, and $(10^6 \cdot 30 \cdot 2^{29})^2 = 2.594 \dots \cdot 10^{32} \leq 30! = 2.652 \dots \cdot 10^{32}$. $\square$

Corollary 2.6. Let $1 \leq m \leq n^2$. If $n \geq N_0$, then $2^{\omega'(m)} \leq n\rho'(m)/1000$. If $n \geq N_1$, then $2^{\omega'(m)} \leq n/10^6$.

Proof. Combine Lemma 2.5 with $(\omega'(m) + 1)\rho'(m) \geq 1$ from Lemma 2.4(a). $\square$

2.4. Two combinatorial lemmas

Lemma 2.7. Let $n \geq 13$ and let O be a set of at least $o(n) + 1$ odd numbers in $[n]$. Then O contains two distinct coprime elements.

Proof. For $i \geq 0$ call $\{6i+1, 6i+3, 6i+5\}$ the i -th block. Two distinct odd numbers in one block differ by 2 or 4, so their greatest common divisor divides 4 and is 1. Suppose that O contains no coprime pair; then each block contains at most one element of O . The blocks meeting $[n]$ are those with $i \leq K = \lfloor (n-1)/6 \rfloor$, and $o(n) = \lfloor (n+3)/6 \rfloor \geq K$. Hence $|O| \geq o(n) + 1 \geq K + 1$ forces $o(n) = K$ and one element of O in every block. The equality $\lfloor (n+3)/6 \rfloor = \lfloor (n-1)/6 \rfloor$ means $n \equiv 1, 2 \pmod{6}$, so the last block meets $[n]$ only in $6K+1$, which therefore lies in O . The element x_0 of O in block 0 is 1, 3 or 5, and the element x_1 in block 1 is 7, 9 or 11 (here $K \geq 2$ as $n \geq 13$). Since 1 is coprime to everything and 5 is coprime to 7, 9 and 11, we get $x_0 = 3$. But 3 is coprime to $6K+1 \neq 3$, a contradiction. $\square$

Lemma 2.8. *Let I be a finite set with a map κ into a linearly ordered set, and for each $i \in I$ let C_i be a finite set with $|C_i| \geq \#\{j \in I : \kappa(j) \leq \kappa(i)\}$. Then there are pairwise distinct elements $w_i \in C_i$ ($i \in I$).*

Proof. Choose the w_i in order of increasing $\kappa(i)$. When i is reached, the values already chosen belong to indices $j \neq i$ with $\kappa(j) \leq \kappa(i)$, and there are fewer than $|C_i|$ of them. $\square$

3. Jackson's theorem and the assembly lemma

Theorem 3.1 (Jackson [5]). *Let H be a bipartite graph with parts X and Y , and let k be an integer with $2 \leq |X| \leq k$ and $|Y| \leq 2k-2$. If every vertex of X has at least k neighbours, then H has a cycle through all vertices of X .*

We include a short proof.⁴

Proof. Two vertices of X have at least $2k - |Y| \geq 2$ common neighbours, which gives a cycle of length 4 through two vertices of X . It suffices to show that a cycle $C = x_0 y_0 x_1 y_1 \cdots x_{t-1} y_{t-1} x_0$ with $x_i \in X$, $y_i \in Y$ and $t < |X|$ can be replaced by such a cycle through $t+1$ vertices of X . Indices are taken modulo t , and $t \leq k-1$.

Fix $x \in X$ not on C . Let $Y' = Y \setminus \{y_0, \dots, y_{t-1}\}$, so $|Y'| = |Y| - t$, and for $v \in X$ let $N'(v)$ be the set of neighbours of v in Y' . Let $I = \{i : x \sim y_i\}$ and $s = |I|$. Suppose that C cannot be enlarged.

- (i) For $i \in I$ the sets $N'(x)$ and $N'(x_{i+1})$ are disjoint: a common element y would give the cycle in which the edge $y_i x_{i+1}$ is replaced by the path $y_i x y x_{i+1}$ (Figure 3(a)).
- (ii) For distinct $i, j \in I$ the sets $N'(x_{i+1})$ and $N'(x_{j+1})$ are disjoint: a common element y would give the cycle

$$x y_i x_i y_{i-1} \cdots y_{j+1} x_{j+1} y x_{i+1} y_{i+1} \cdots x_j y_j x,$$

which runs backwards along C from x_i to x_{j+1} and forwards from x_{i+1} to x_j (Figure 3(b)).

The vertices x_{i+1} , $i \in I$, are distinct. The vertex x has exactly s neighbours on C , so $|N'(x)| \geq k-s$, and $|N'(x_{i+1})| \geq k-t \geq 1$. If $s \geq 1$, the disjoint sets in (i) and (ii) give

$$(k-s) + s(k-t) \leq |Y'| \leq 2k-2-t.$$

As $s(k-t) \geq (k-t) + (s-1)$, the left side is at least $2k-t-1$, a contradiction. If $s = 0$, then x has no neighbour on C . Choose y' adjacent to x and x_0 , and $y'' \neq y'$ adjacent to x and x_1 ; both exist because any two vertices of X have at least two common neighbours, and both lie off C . Replacing $x_0 y_0 x_1$ by $x_0 y' x y'' x_1$ enlarges C (Figure 3(c)). $\square$

In our applications the vertices of X are paths in the coprime graph, and the vertices of Y are single integers that join them.

⁴Jackson's theorem belongs to the family of degree conditions for long cycles that started with Dirac's theorem on Hamiltonian cycles. Erdős and Sárközy used it in [4], and it is the engine of all constructions in this paper. The argument given here, a maximal cycle together with one counting inequality, is the one formalized in Lean.

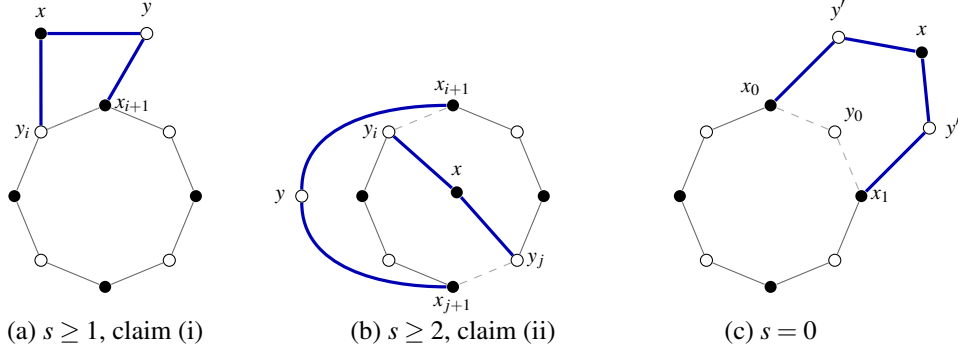


Figure 3: Enlarging a cycle in the proof of Theorem 3.1. Filled vertices lie in X , open ones in Y ; thick edges are new and dashed edges are removed.

Lemma 3.2. *Let A be a finite set of positive integers and k an integer. Let X be a finite index set and, for $c \in X$, let $P_c = (v_1^c, \dots, v_{m_c}^c)$ with $m_c \geq 1$ be a sequence of distinct elements of A in which consecutive terms are coprime; let the P_c be pairwise disjoint. Let $B \subseteq A$ be disjoint from all P_c , and call $\beta \in B$ attached to P_c if β is coprime to v_1^c and to $v_{m_c}^c$. If $2 \leq |X| \leq k$, $|B| \leq 2k - 2$ and every P_c has at least k attached elements of B , then $G(A)$ contains a cycle of length $\sum_{c \in X} m_c + |X|$.*

Proof. Apply Theorem 3.1 to the bipartite graph H on $X \cup B$ whose edges are the attachments. It yields a cycle $c_0 \beta_0 c_1 \beta_1 \dots c_{t-1} \beta_{t-1} c_0$ with $\{c_0, \dots, c_{t-1}\} = X$. Replacing each c_i by P_{c_i} gives the closed sequence $P_{c_0} \beta_0 P_{c_1} \beta_1 \dots P_{c_{t-1}} \beta_{t-1}$ of distinct elements of A ; consecutive terms are coprime inside each P_c , and β_i is coprime to the last term of P_{c_i} and to the first term of $P_{c_{i+1}}$. Its length is $\sum_c m_c + t \geq 4$. Writing Φ for this expansion and ℓ for the length of a cycle, the triangle

$$\begin{array}{ccc} \{\text{cycles of } H \text{ through all of } X\} & \xrightarrow{\Phi} & \{\text{cycles of } G(A)\} \\ & \searrow & \downarrow \ell \\ & \sum_c m_c + |X| & \mathbb{N} \end{array} \quad (4)$$

commutes, where the diagonal map is constant. $\square$

4. How many integers have small ρ'

For a finite set $\mathcal{T}$ of primes let $\varphi_{\mathcal{T}}(z) = \prod_{p \in \mathcal{T}, p|z} (1 - 1/p)$, and for $m \geq 0$ let $g_m(p) = (p/(p-1))^m - 1 \geq 0$.

Lemma 4.1 (moment bound). *For every finite set $\mathcal{T}$ of primes, all integers $n, m \geq 0$ and every $x > 0$,*

$$\#\{z \in [n] : \varphi_{\mathcal{T}}(z) \leq x\} \leq nx^m \prod_{p \in \mathcal{T}} \left(1 + \frac{g_m(p)}{p}\right).$$

Proof. If $\varphi_{\mathcal{T}}(z) \leq x$, then $1 \leq (x/\varphi_{\mathcal{T}}(z))^m = x^m \prod_{p \in \mathcal{T}, p|z} (1 + g_m(p))$. Expanding the product and summing over $z \leq n$,

$$\sum_{z \leq n} \prod_{p \in \mathcal{T}, p|z} (1 + g_m(p)) = \sum_{D \subseteq \mathcal{T}} \left\lfloor \frac{n}{\prod_{p \in D} p} \right\rfloor \prod_{p \in D} g_m(p) \leq n \prod_{p \in \mathcal{T}} \left(1 + \frac{g_m(p)}{p}\right). \quad \square$$

This is Rankin's method.⁵ The Euler products are controlled as follows. For $s \geq 5$ let

$$F_s(m) = \prod_{s \leq p < 200} \left(1 + \frac{g_m(p)}{p}\right) \in \mathbb{Q}.$$

⁵R. A. Rankin [8] introduced the device of bounding the number of integers with a multiplicative property by a weighted sum with an adjustable exponent, in his work on large gaps between consecutive primes. Here the exponent m is chosen from a finite menu, separately for each signature in Theorem 4.4.

Lemma 4.2. *If $2m+2 \leq 200$ and $\mathcal{T}$ is a finite set of primes $p \geq s$, then $\prod_{p \in \mathcal{T}} (1 + g_m(p)/p) \leq F_s(m)/(1 - 2m/199)$.*

Proof. For $p \geq 2m+2$ put $v = 1/(p-1)$. Bernoulli's inequality gives $1 - mv \leq (1 - v)^m$, and $(1 + v)^m(1 - v)^m \leq 1$, so $(1 + v)^m \leq 1/(1 - mv)$ and $g_m(p) \leq mv/(1 - mv) = m/(p-1-m) \leq 2m/p$. For any finite set of integers $p \geq 200$ we have $\sum 1/p^2 \leq \sum (1/(p-1) - 1/p) \leq 1/199$, and for $a_p \geq 0$ the inequality $\prod (1 + a_p) \cdot (1 - \sum a_p) \leq 1$ holds by induction. With $a_p = 2m/p^2$ the primes $p \geq 200$ of $\mathcal{T}$ contribute at most $1/(1 - 2m/199)$, and the primes below 200 at most $F_s(m)$, all factors being at least 1. $\square$

Call an odd number z *hard* if $\rho'(z) < 14/25$ and *easy* if $\rho'(z) \geq 3/4$.

Corollary 4.3. *For every $n \geq 1$:*

- (a) $\#\{z \in [n] : \rho'(z) \leq x\} \leq xn/560$ for $0 < x \leq 14/25$; in particular at most $n/1000$ integers in $[n]$ are hard;
- (b) at most $11(\lfloor n/70 \rfloor + 1) + n/1000$ odd integers in $[n]$ are not easy.

Proof. (a) For $z \leq n$ we have $\rho'(z) = \phi_{\mathcal{T}}(z)$ with $\mathcal{T}$ the primes in $[5, n]$. The Lean kernel verifies the rational inequality

$$(14/25)^{37} F_5(37) \leq \frac{1}{1000} (1 - 74/199). \quad (5)$$

By Lemmas 4.1 and 4.2 with $m = 37$, and $x^{37} \leq x(14/25)^{36}$, the count is at most

$$nx \cdot \frac{25}{14} \cdot \frac{(14/25)^{37} F_5(37)}{1 - 74/199} \leq \frac{25}{14} \cdot \frac{nx}{1000} = \frac{nx}{560}.$$

(b) Among any 70 consecutive integers exactly 11 are odd and divisible by 5 or 7, so $[n]$ contains at most $11(\lfloor n/70 \rfloor + 1)$ of them. For the other odd $z \leq n$, $\rho'(z) = \phi_{\mathcal{T}}(z)$ with $\mathcal{T}$ the primes in $[11, n]$, and the kernel verifies $(3/4)^{54} F_{11}(54) \leq \frac{1}{1000} (1 - 108/199)$; Lemmas 4.1 and 4.2 with $m = 54$ bound their number by $n/1000$. $\square$

The second estimate concerns residue classes and finer thresholds. Let $P_{19} = \{5, 7, 11, 13, 17, 19\}$. For $S \subseteq P_{19}$ put

$$\rho'(S) = \prod_{p \in S} \left(1 - \frac{1}{p}\right), \quad \text{dens}(S) = \frac{\prod_{p \in P_{19} \setminus S} (1 - 1/p)}{\prod_{p \in S} p},$$

the density of the integers whose set of prime divisors in P_{19} is exactly S . Let

$$\mathcal{M} = \{0, 3, 6, 10, 15, 20, 25, 30, 40, 50, 60, 70, 85, 99\},$$

and let $\hat{F}_m$ ($m \in \mathcal{M}$) be the rationals of Table 1, which satisfy $F_{23}(m) \leq \hat{F}_m(1 - 2m/199)$. For $\theta' > 0$ define

$$\text{cert}(\theta') = \sum_{S \subseteq P_{19}} \text{dens}(S) \min_{m \in \mathcal{M}} \left(\frac{\theta'}{\rho'(S)} \right)^m \hat{F}_m.$$

For a symmetric R and $\theta > 0$ put $\text{Bad}_R(n, \theta) = \#\{z \in [n] : z \bmod 6 \in R, \rho'(z) < \theta\}$.

Theorem 4.4 (density certificates). *Let θ, θ', β be rationals with $0 < \theta \leq \theta'(1 - 10^{-4})$ and $\text{cert}(\theta') + 7 \cdot 10^{-6} \leq \beta$. Then there is N such that for every $n \geq N$ and every nonempty symmetric $R \subseteq \mathbb{Z}/6\mathbb{Z}$,*

$$\text{Bad}_R(n, \theta) \leq \frac{n}{6} |R| \beta.$$

The hypotheses hold for the sixteen triples of Table 2.

Table 1: The constants $\hat{F}_m$ ($\hat{F}_0 = 1$). The middle column is $F_{23}(m)/(1 - 2m/199)$ rounded to 8 decimals; the inequality $F_{23}(m) \leq \hat{F}_m(1 - 2m/199)$ is checked exactly by the Lean kernel.

m	$F_{23}(m)/(1 - 2m/199)$	$\hat{F}_m$
3	1.06276861	1.062769
6	1.13338522	1.133386
10	1.24218851	1.242189
15	1.40781548	1.407816
20	1.61704135	1.617042
25	1.88618635	1.886187
30	2.23941115	2.239412
40	3.36498270	3.364983
50	5.62632408	5.626325
60	10.80856280	10.808563
70	24.90239418	24.902395
85	146.58258692	146.582587
99	15323.30517473	15323.305175

Proof. Fix n and R , and let $Y = 10^{10}$ and $\eta = 10^{-4}$. For $z \leq n$ let S_z be the set of primes of P_{19} dividing z , let $\mathcal{T}$ be the primes in $[23, Y]$ and $\mathcal{B}$ the primes in $(Y, n]$. Splitting the primes of $\mathcal{P}(z)$ into the three ranges, that is, $\sigma(z) = (S_z, \mathcal{P}(z) \cap \mathcal{T}, \mathcal{P}(z) \cap \mathcal{B})$, gives the commutative diagram

$$\begin{array}{ccc} [n] & \xrightarrow{\sigma} & 2^{P_{19}} \times 2^{\mathcal{T}} \times 2^{\mathcal{B}} \\ \mathcal{P} \downarrow & & \downarrow \pi \\ 2^{\{p \geq 5\}} & \xrightarrow{\pi_1} & (0, 1] \end{array} \quad (6)$$

in which $\pi_1(Q) = \prod_{p \in Q} (1 - 1/p)$ and π sends a triple of disjoint sets to π_1 of their union; hence $\rho'(z) = \rho'(S_z) \varphi_{\mathcal{T}}(z) \varphi_{\mathcal{B}}(z)$.

Large primes. We have $\varphi_{\mathcal{B}}(z) \geq 1 - \sum_{p \in \mathcal{B}, p|z} 1/p$. The number of $z \leq n$ with $\sum_{p \in \mathcal{B}, p|z} 1/p > \eta$ is at most

$$\frac{1}{\eta} \sum_{z \leq n} \sum_{p \in \mathcal{B}, p|z} \frac{1}{p} \leq \frac{1}{\eta} \sum_{p > Y} \frac{n}{p^2} \leq \frac{n}{\eta Y} = \frac{n}{10^6}.$$

For every other z with $\rho'(z) < \theta$ we get $\rho'(S_z) \varphi_{\mathcal{T}}(z) (1 - \eta) < \theta \leq \theta' (1 - \eta)$, so $\varphi_{\mathcal{T}}(z) < \theta' / \rho'(S_z)$.

Middle primes. Fix $S \subseteq P_{19}$ and let $Z_S = \{z \in [n] : z \bmod 6 \in R, S_z = S\}$. For $D \subseteq \mathcal{T}$ the elements of Z_S divisible by $\prod_{p \in D} p$ are the numbers $(\prod_{p \in S \cup D} p) w$ with w counted by $C_R(\lfloor n / \prod_{p \in S \cup D} p \rfloor; P_{19} \setminus S)$; this is the argument of diagram (2), as the multiplier is $\equiv \pm 1 \pmod{6}$. By Lemma 2.2 this number is at most

$$\frac{n}{6} |R| \text{dens}(S) \prod_{p \in D} \frac{1}{p} + |R| 2^6.$$

The proof of Lemma 4.1, run inside Z_S , gives for every m and $t > 0$

$$\#\{z \in Z_S : \varphi_{\mathcal{T}}(z) \leq t\} \leq t^m \left(\frac{n}{6} |R| \text{dens}(S) \prod_{p \in \mathcal{T}} \left(1 + \frac{g_m(p)}{p} \right) + |R| 2^6 \prod_{p \in \mathcal{T}} (1 + g_m(p)) \right).$$

By Lemma 4.2, $\prod_{p \in \mathcal{T}} (1 + g_m(p)/p) \leq F_{23}(m)/(1 - 2m/199) \leq \hat{F}_m$ for $m \in \mathcal{M}$. Take $t = \theta' / \rho'(S)$ and let $m_S \in \mathcal{M}$ minimize $t^{m_S} \hat{F}_{m_S}$.

Summation. Summing over S ,

$$\text{Bad}_R(n, \theta) \leq \frac{n}{10^6} + \frac{n}{6} |R| \text{cert}(\theta') + K, \quad K = \sum_S \left(\frac{\theta'}{\rho'(S)} \right)^{m_S} 6 \cdot 2^6 \prod_{p \in \mathcal{T}} (1 + g_{m_S}(p)),$$

Table 2: The sixteen density certificates: thresholds θ , inflated thresholds θ' , the value $\text{cert}(\theta') + 7 \cdot 10^{-6}$ rounded up to 6 decimals, and the certified bound β .

θ	θ'	$\text{cert}(\theta') + 7 \cdot 10^{-6}$	β
0.550	0.5502	0.000218	0.0003
0.600	0.6002	0.003403	0.0035
0.625	0.6252	0.007914	0.0080
0.650	0.6502	0.017781	0.0178
0.675	0.6752	0.035261	0.0353
0.700	0.7002	0.050146	0.0502
0.725	0.7252	0.074385	0.0744
0.750	0.7502	0.114311	0.1144
0.775	0.7752	0.209736	0.2098
0.800	0.8002	0.248244	0.2483
0.825	0.8252	0.303389	0.3034
0.850	0.8502	0.345598	0.3457
0.875	0.8752	0.393815	0.3939
0.900	0.9002	0.463673	0.4637
0.925	0.9252	0.538623	0.5387
0.950	0.9502	0.710685	0.7107

where K does not depend on n . Put $N = \lceil 6 \cdot 10^6 K \rceil$. For $n \geq N$ we have $K \leq n/(6 \cdot 10^6)$, and $n/10^6 + n/(6 \cdot 10^6) = (n/6) \cdot 7 \cdot 10^{-6} \leq (n/6)|R| \cdot 7 \cdot 10^{-6}$. Hence $\text{Bad}_R(n, \theta) \leq (n/6)|R|(\text{cert}(\theta') + 7 \cdot 10^{-6}) \leq (n/6)|R|\beta$.

The sixteen certificates. For each row of Table 2, $\text{cert}(\theta')$ is a sum of 64 rationals, each a minimum of 14 explicit rationals. The Lean kernel evaluates $\text{cert}(\theta') + 7 \cdot 10^{-6} \leq \beta$ exactly. Write $c(\theta', \beta) = \text{cert}(\theta') + 7 \cdot 10^{-6} - \beta$ for the rational function and $c_{\mathbb{R}}$ for the same formula over $\mathbb{R}$. Since $\iota : \mathbb{Q} \rightarrow \mathbb{R}$ is an embedding of ordered fields, the square

$$\begin{array}{ccc}
 \mathbb{Q}^2 & \xrightarrow{c} & \mathbb{Q} \\
 \iota \times \iota \downarrow & & \downarrow \iota \\
 \mathbb{R}^2 & \xrightarrow{c_{\mathbb{R}}} & \mathbb{R}
 \end{array} \tag{7}$$

commutes, because c is built from field operations and minima, which ι preserves. As ι also preserves signs, the sixteen inequalities proved over $\mathbb{Q}$ hold over $\mathbb{R}$. $\square$

The number N is finite and determined by the proof; its size plays no role. Figure 4 compares the certified values with the actual proportions up to 10^8 .

5. Triangles

Theorem 5.1. *If $n \geq 13$ and $A \subseteq [n]$ with $|A| > T(n)$, then $G(A)$ contains a triangle.*

Proof. Let O be the set of odd elements of A and e the number of even integers in $[n]$ missing from A ; by Lemma 2.1, $|O| \geq o(n) + 1 + e$. Use the blocks $\{6i + 1, 6i + 3, 6i + 5\}$ of Lemma 2.7. Odd numbers in one block are pairwise coprime, and each is coprime to $6i + 2$ and to $6i + 4$ (Figure 5). Suppose that $G(A)$ has no triangle. Then every block contains at most two elements of O , and if it contains two, then neither $6i + 2$ nor $6i + 4$ lies in A .

Let $F \subseteq O$ consist of the smallest element of O in each block that meets O , and let $P = O \setminus F$. Then $|F| \leq \lfloor (n-1)/6 \rfloor + 1$, and P has at most one element in each block. For $v \in P$ in block i the number

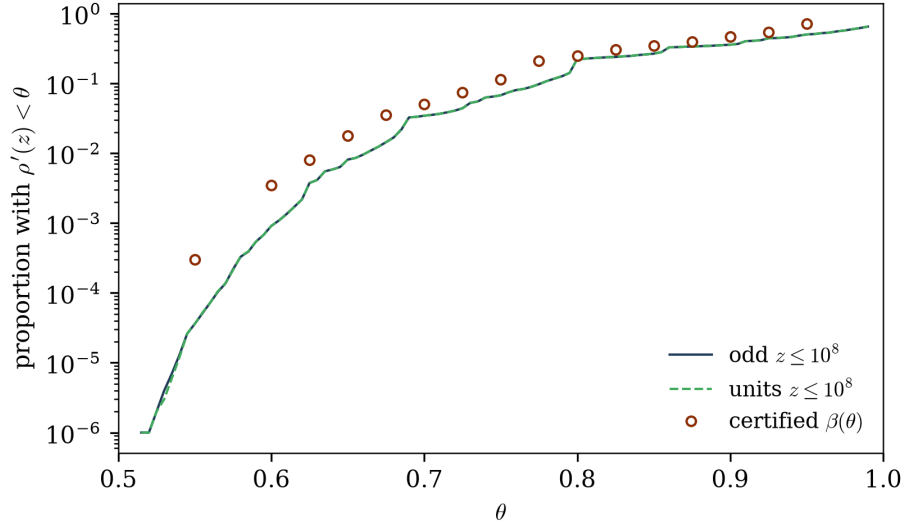


Figure 4: Proportion of odd numbers and of units $z \leq 10^8$ with $\rho'(z) < \theta$, against the certified bounds $\beta(\theta)$ of Table 2 (logarithmic scale).

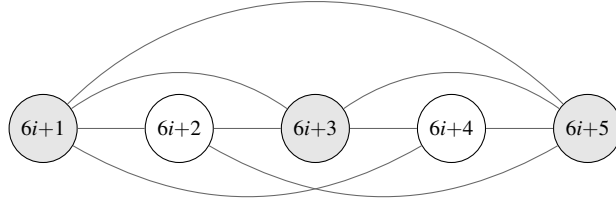


Figure 5: The coprime pairs inside $\{6i+1, \dots, 6i+5\}$ that are used in Theorem 5.1: all pairs of odd elements and all pairs of an odd and an even element.

$6i+2 \leq n$ is missing from A , and these numbers are distinct, so $|P| \leq e$. Hence

$$o(n) + 1 + e \leq |O| = |F| + |P| \leq \lfloor (n-1)/6 \rfloor + 1 + e.$$

Since $o(n) \geq \lfloor (n-1)/6 \rfloor$, equality holds throughout: $o(n) = \lfloor (n-1)/6 \rfloor$, so $n \equiv 1, 2 \pmod{6}$, and $|P| = e$. For $v \in P$ in block i we have $v \leq n$ and $v \geq 6i+3$, so $6i+4 \leq n$ by the congruence of n , and $6i+4$ is missing from A as well. The numbers $6i+2$ and $6i+4$ obtained from all $v \in P$ are $2|P|$ distinct missing even numbers, so $2|P| \leq e = |P|$ and $e = 0$. In particular $2 \in A$. By Lemma 2.7 there are distinct coprime $a, b \in O$, and $a, b, 2$ is a triangle. $\square$

6. Sets that miss few even numbers prime to 3

Theorem 6.1. *Let $n \geq N_0$ and $A \subseteq [n]$ with $|A| > T(n)$ and $70d \leq n$. Then $G(A)$ contains a cycle of length $2l+1$ for every integer l with $1 \leq l \leq o(n)$.*

Since $\lfloor n/6 \rfloor \leq o(n)$, this gives Theorem 1.1 for these sets. Throughout this section $n \geq N_0$, A is as in Theorem 6.1, and all numbers lie in $[n]$. The deficit of z is $a(z) = |N^*(n, z) \setminus A| \leq d$, and z is *undamaged* if $a(z) \leq (n/8)\rho'(z)$.

Lemma 6.2. *At most d integers in $[n]$ are damaged.*

Proof. If $d = 0$ every integer is undamaged. Otherwise a damaged z satisfies $(n/8)\rho'(z) < a(z) \leq d$, so $\rho'(z) < x = 8d/n \leq 8/70 < 14/25$, and by Corollary 4.3(a) there are at most $xn/560 = d/70 \leq d$ such z . $\square$

Lemma 6.3. *If y is odd and easy and z is odd and undamaged, then $|N^*(n, \text{lcm}(y, z)) \cap A| \geq 2 + \rho'(z)n/280$.*

Proof. Put $m = \text{lcm}(y, z)$ and $K = \omega'(m)$; then m is odd and $m \leq n^2$. By Corollary 2.3, $|N^*(n, m)| \geq (n/3)\rho'(m) - 2^K$, and by Lemma 2.4, $\rho'(m) \geq \frac{3}{4}\rho'(z)$. The elements of $N^*(n, m)$ missing from A lie in $N^*(n, z) \setminus A$, so there are at most $a(z) \leq (n/8)\rho'(z)$ of them. By Corollary 2.6 and Lemma 2.4(b), $2^K \leq n\rho'(m)/1000 \leq n\rho'(z)/1000$; in particular $n\rho'(z) \geq 1000$. Hence

$$|N^*(n, m) \cap A| \geq n\rho'(z) \left(\frac{1}{4} - \frac{1}{8} - \frac{1}{1000} \right) \geq 2 + \frac{n\rho'(z)}{280}. \quad \square$$

Lemma 6.4. *Let p, q be odd with $\rho'(\text{lcm}(p, q)) \geq 14/25$, and let W be a set with $|W| \leq n/500 + 2$. Then $N^*(n, \text{lcm}(p, q)) \setminus ((E^*(n) \setminus A) \cup W)$ has at least $\lfloor n/6 \rfloor + 2$ elements.*

Proof. By Corollaries 2.3 and 2.6, the set has at least $\frac{n}{3} \cdot \frac{14}{25} - \frac{n}{1000} - d - \frac{n}{500} - 2 \geq n \left(\frac{14}{75} - \frac{1}{1000} - \frac{1}{70} - \frac{1}{500} \right) - 2$ elements. The bracket exceeds $\frac{1}{6} + 0.0027$, and $0.0027n \geq 4$. $\square$

Lemma 6.5. *Let O'' be a finite set of odd numbers, H its hard and E its easy elements, with $|E| \geq 2|H| + 2$, and let $l \geq 4$ with $l - 1 \leq |O''|$. Then there is $X_1 \subseteq O''$ with $|X_1| = l - 1$ whose sets H_1 of hard and E_1 of easy elements satisfy $|E_1| \geq 2|H_1| + 2$, $l \geq 2|H_1| + 4$ and $|H_1| \leq |H|$.*

Proof. Let $N = O'' \setminus H$, so $E \subseteq N$. If $|N| \geq l - 1$, take $X_1 \subseteq N$ of size $l - 1$ containing two elements of E ; then $H_1 = \emptyset$. Otherwise take $X_1 = N \cup H'$ with $H' \subseteq H$ of size $l - 1 - |N| \geq 1$. Then $H_1 = H'$ and $E_1 = E$, so $|E_1| \geq 2|H| + 2 \geq 2|H_1| + 2$, and $l - 1 = |H_1| + |N| \geq |H_1| + |E| \geq 3|H_1| + 2$, whence $l \geq 2|H_1| + 4$. $\square$

Proof of Theorem 6.1. For $l = 1$ apply Theorem 5.1; let $l \geq 2$. By Lemma 2.1 with $D = E^*(n) \setminus A$, the set O of odd elements of A has at least $o(n) + 1 + d$ elements, and by Lemma 6.2 the set O' of its undamaged elements has at least $o(n) + 1$. By Lemma 2.7 there are distinct coprime $a, b \in O'$. Let $O'' = O' \setminus \{a, b\}$, so $|O''| \geq o(n) - 1$, and let H and E be its sets of hard and easy elements. By Corollary 4.3, $|H| \leq n/1000$ and $|O'' \setminus E| \leq 11n/70 + 11 + n/1000$. With $o(n) \geq n/6 - 1$ this gives

$$|E| \geq \frac{n}{105} - \frac{n}{1000} - 13 \geq \frac{n}{500} + 2 \geq 2|H| + 2.$$

$l = 2$. Take $y \in E$. By Lemma 6.3 the sets $N^*(n, \text{lcm}(y, a)) \cap A$ and $N^*(n, \text{lcm}(y, b)) \cap A$ have at least two elements, so they contain distinct w_a and w_b . Then a, b, w_b, y, w_a is a cycle of length 5.

$l = 3$. Take distinct $y_a, y_b \in E$. By Lemma 6.3, applied three times (y_b is undamaged), each of the sets $N^*(n, \text{lcm}(y_a, a)) \cap A$, $N^*(n, \text{lcm}(y_b, b)) \cap A$ and $N^*(n, \text{lcm}(y_a, y_b)) \cap A$ has at least three elements, so they contain distinct w_0, w_1, w_2 . Then $y_a, w_0, a, b, w_1, y_b, w_2$ is a cycle of length 7.

$l \geq 4$. Choose X_1 by Lemma 6.5, with hard part H_1 and easy part E_1 . Its elements are odd, undamaged and different from a, b . Consider the set of slots

$$\Sigma = \{(a, 0), (b, 1)\} \cup H_1 \times \{0, 1\}, \quad |\Sigma| = 2|H_1| + 2,$$

and call the first coordinate of a slot its owner. Since $|E_1| \geq |\Sigma|$, there is an injective choice of partners $\pi(\sigma) \in E_1$. For a slot σ with owner z let $C_\sigma = N^*(n, \text{lcm}(\pi(\sigma), z)) \cap A$, and order the slots by ρ' of their owners. The slots whose owner z' has $\rho'(z') \leq \rho'(z)$ number at most $2 + 2\#\{c \in H_1 : \rho'(c) \leq$

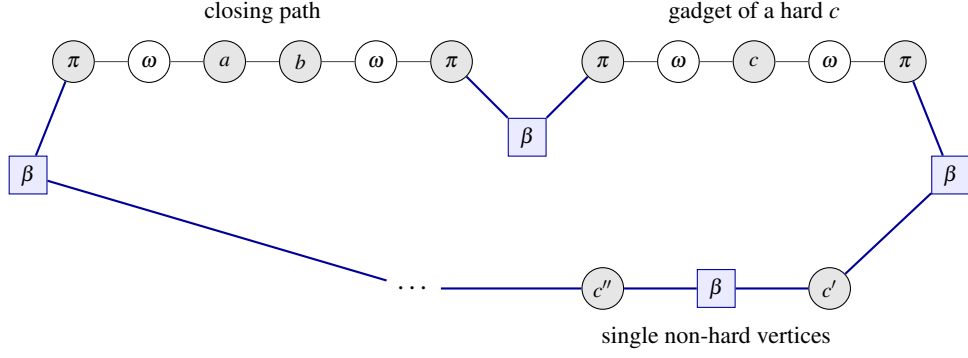


Figure 6: The cycle in the proof of Theorem 6.1. Grey vertices are odd elements of A , white vertices are connectors from $E^*(n) \cap A$, and the boxed vertices β are taken from the pool by Jackson's theorem.

$\rho'(z) \leq 2 + \rho'(z)n/280$, by Corollary 4.3(a) when $\rho'(z) \leq 14/25$ and by $|H_1| \leq n/1000$ otherwise. By Lemmas 6.3 and 2.8 there are distinct connectors $\omega(\sigma) \in C_\sigma$. Form the sequences

$$\begin{aligned} P_a &= (\pi(a, 0), \omega(a, 0), a, b, \omega(b, 1), \pi(b, 1)), \\ P_c &= (\pi(c, 0), \omega(c, 0), c, \omega(c, 1), \pi(c, 1)) \quad (c \in H_1), \\ P_c &= (c) \quad (c \in R), \quad R = X_1 \setminus (H_1 \cup \pi(\Sigma)), \end{aligned}$$

indexed by $X = \{a\} \cup H_1 \cup R$ (Figure 6). Consecutive terms are coprime by the choice of the connectors and since a, b are coprime. The sequences are pairwise disjoint: the odd terms are the distinct elements $a, b, \pi(\sigma)$ and the elements of $H_1 \cup R$, and the even terms are the distinct connectors. We have $|R| = l - 1 - |H_1| - |\Sigma| = l - 3 - 3|H_1|$, hence $|X| = 1 + |H_1| + |R| = l - 2 - 2|H_1| \geq 2$ and $|X| \leq l \leq o(n) \leq k$, where $k = \lfloor n/6 \rfloor + 2$.

The pool is $B = (E^*(n) \cap A) \setminus \omega(\Sigma)$; it is disjoint from all P_c and $|B| \leq |E^*(n)| = \lfloor n/2 \rfloor - \lfloor n/6 \rfloor \leq 2k - 2$. The ends of P_a and P_c ($c \in H_1$) are two easy numbers, with ρ' of their least common multiple at least $(3/4)^2 > 14/25$; the single vertices $c \in R$ are not hard. Lemma 6.4 with $W = \omega(\Sigma)$, where $|W| = 2|H_1| + 2 \leq n/500 + 2$, shows that each sequence has at least k attached elements in B . Lemma 3.2 gives a cycle of length

$$6 + 5|H_1| + |R| + |X| = 7 + 6|H_1| + 2|R| = 2l + 1. \quad \square$$

7. Two constructions for sets that miss many even numbers prime to 3

In this section and the next $n \geq N_1$. For $A \subseteq [n]$ and $t \in \mathbb{R}$ let

$$\mathcal{O}_A(t) = \{z \in A : z \text{ odd}, \rho'(z) \geq t\}, \quad \mathcal{U}_A(t) = \{z \in A \cap U(n) : \rho'(z) \geq t\}.$$

Lemma 7.1. *Let $1 \leq m \leq n^2$. For odd m , $|N^*(n, m)| \geq (n/3)\rho'(m) - n/10^6$. If R is symmetric and nonempty, and m and R satisfy the hypothesis of Corollary 2.3, then $\#\{w \in [n] : w \bmod 6 \in R, \gcd(w, m) = 1\} \geq (n/6)|R|\rho'(m) - |R|n/10^6$.*

Proof. Corollaries 2.3 and 2.6. $\square$

Proposition 7.2 (unit construction). *Let $A \subseteq [n]$, let $2 \leq l \leq k$ be integers and θ_T, r_1, t_2 reals with $r_1 \geq 0$ and $\theta_T \leq r_1 t_2$. Suppose that*

- (T1) $|\mathcal{U}_A(\theta_T)| \geq l + 1$;
- (T2) $\mathcal{U}_A(r_1) \neq \emptyset$;
- (T3) $2 + 2n/10^6 + |U(n) \setminus A| + \#\{z \in U(n) : \rho'(z) < t_2\} \leq (n/3)r_1$;

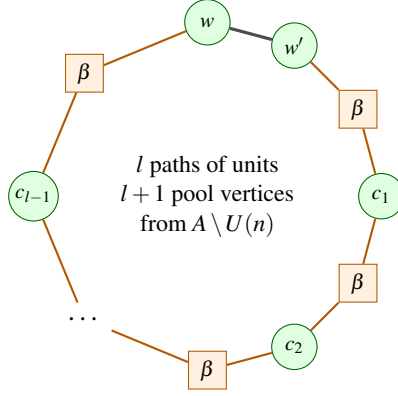


Figure 7: The unit construction: the path w, w' and $l - 1$ single units of large ρ' , joined by pool vertices from $M(n) \cap A$.

$$(T4) \quad k + \mu_T + 4n/10^6 \leq (2n/3)\theta_T;$$

$$(T5) \quad T(n) + 2 \leq 2k + \mu_T.$$

Then $G(A)$ contains a cycle of length $2l + 1$.

Proof. Take $w \in \mathcal{U}_A(r_1)$. By Lemma 7.1 with $R = \{1, 5\}$ and $m = w$, at least $(n/3)\rho'(w) - 2n/10^6 \geq (n/3)r_1 - 2n/10^6$ units of $[n]$ are coprime to w . Removing those missing from A and those with $\rho' < t_2$ leaves at least two by (T3), so there is $w' \neq w$ in $\mathcal{U}_A(t_2)$ coprime to w . By (T1) there is a set $S \subseteq \mathcal{U}_A(\theta_T) \setminus \{w, w'\}$ of $l - 1$ units. Apply Lemma 3.2 to the sequences (w, w') and (c) , $c \in S$, indexed by $X = \{w\} \cup S$, with pool $B = M(n) \cap A$ (Figure 7). Here $|X| = l \leq k$ and $|B| = T(n) - \mu_T \leq 2k - 2$ by (T5). The sequences consist of units, so they avoid B . For the ends p, q of a sequence we have $\rho'(\text{lcm}(p, q)) \geq \theta_T$: for (w, w') by Lemma 2.4(c), $\rho'(\text{lcm}(w, w')) \geq r_1 t_2 \geq \theta_T$, and for (c) by the choice of S . As $\text{lcm}(p, q)$ is prime to 6, Lemma 7.1 with $R = \{0, 2, 3, 4\}$ gives at least $(2n/3)\theta_T - 4n/10^6$ elements of $M(n)$ coprime to p and q ; at most μ_T of them are missing from A , so by (T4) each sequence has at least k attached elements. The cycle has length $(2 + (l - 1)) + l = 2l + 1$. $\square$

Proposition 7.3 (giant-path construction). *Let $A \subseteq [n]$, let $l, k \geq 2$ be integers and $s, t_1, \theta, \theta_w, s_w$ reals with $t_1, \theta_w \geq 0$, $s_w \leq s$, and $s_w \leq \theta_w$ if $k < l$. Suppose that*

$$(H1) \quad |\mathcal{O}_A(s)| \geq o(n) + 1;$$

$$(H2) \quad |\mathcal{O}_A(t_1)| \geq 3;$$

$$(H3) \quad \theta \leq t_1 s;$$

$$(H4) \quad \text{if } k < l, \text{ then } |\mathcal{U}_A(\theta_w)| \geq l - k + 2;$$

$$(H5) \quad d + 1 + n/10^6 \leq (n/3)t_1 s_w;$$

$$(H6) \quad \text{if } k < l, \text{ then } (l - k - 1) + \mu_6 + n/10^6 \leq (n/6)\theta_w s_w;$$

$$(H7) \quad |\mathcal{O}_A(\theta)| \geq l + 1;$$

$$(H8) \quad |E^*(n)| + 1 \leq 2k + d;$$

$$(H9) \quad k + d + 1 + n/10^6 \leq (n/3)\theta.$$

Then $G(A)$ contains a cycle of length $2l + 1$.

Proof. By (H1) and Lemma 2.7 there are distinct coprime odd $x, x' \in \mathcal{O}_A(s)$. One of them is not a multiple of 3 (two multiples of 3 are not coprime) and is therefore a unit; call it w° and the other g_2 . By (H2) there is $g_1 \in \mathcal{O}_A(t_1) \setminus \{w^\circ, g_2\}$. By Lemma 2.4(c), $\rho'(\text{lcm}(g_1, g_2)) \geq t_1 s \geq \theta$ by (H3). For every odd p with $\rho'(p) \geq s_w$ we have $\rho'(\text{lcm}(g_1, p)) \geq t_1 s_w$, and Lemma 7.1 with (H5) gives

$$|N^*(n, \text{lcm}(g_1, p)) \cap A| \geq (n/3)t_1 s_w - n/10^6 - d \geq 1. \quad (*)$$

Likewise, for odd p, q with $\rho'(\text{lcm}(p, q)) \geq \theta$, the set $(E^*(n) \cap A) \setminus \{e\}$, for any e , contains at least $(n/3)\theta - n/10^6 - d - 1 \geq k$ elements coprime to p and q , by (H9).

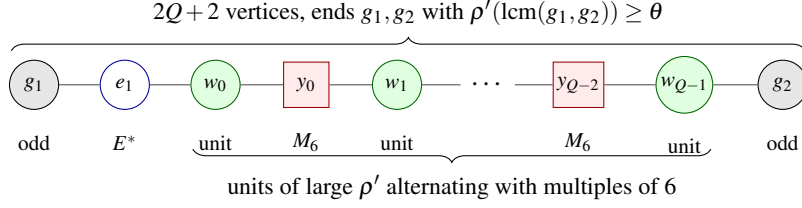


Figure 8: The giant path of Proposition 7.3. It absorbs the $Q = l - k$ odd vertices that the pool $E^*(n) \cap A$ cannot carry when $k < l$.

$l = 2$. By $(*)$ there is $e_1 \in N^*(n, \text{lcm}(g_1, w_0)) \cap A$, and there is $\beta \in E^*(n) \cap A$, $\beta \neq e_1$, coprime to g_1 and g_2 . Then $g_1, e_1, w_0, g_2, \beta$ is a cycle of length 5.

$l \geq 3$. Let $Q = 1$ if $l \leq k$ and $Q = l - k$ otherwise; then $1 \leq Q \leq l - 2$ and $l - Q \leq k$. Put $w_{Q-1} = w_0$. If $Q \geq 2$, then $k < l$ and $Q - 1 = l - k - 1$, and (H4) provides distinct units $w_0, \dots, w_{Q-2} \in \mathcal{U}_A(\theta_w) \setminus \{g_1, g_2, w_0\}$. All w_i satisfy $\rho'(w_i) \geq s_w$. By $(*)$ there is $e_1 \in N^*(n, \text{lcm}(g_1, w_0)) \cap A$. If $Q \geq 2$, then for $i \leq Q - 2$ the number $\text{lcm}(w_i, w_{i+1})$ is prime to 6 and has $\rho' \geq \theta_w s_w$; Lemma 7.1 with $R = \{0\}$ gives at least $(n/6)\theta_w s_w - n/10^6 - \mu_6 \geq Q - 1$ elements of $M_6(n) \cap A$ coprime to $w_i w_{i+1}$, by (H6), and Lemma 2.8 yields distinct such elements y_i . The *giant path*

$$\Gamma = (g_1, e_1, w_0, y_0, w_1, y_1, \dots, y_{Q-2}, w_{Q-1}, g_2)$$

has $2Q + 2$ distinct terms, consecutive terms being coprime (Figure 8). Its odd terms form a set W_{odd} of at most $Q + 2$ elements, so by (H7) there is $S \subseteq \mathcal{O}_A(\theta) \setminus W_{\text{odd}}$ with $|S| = l - 1 - Q$.

Apply Lemma 3.2 to Γ and the sequences (c) , $c \in S$, indexed by $X = \{g_1\} \cup S$, with pool $B = (E^*(n) \cap A) \setminus \{e_1\}$ and the integer k . Here $2 \leq |X| = l - Q \leq k$, and $|B| = |E^*(n)| - d - 1 \leq 2k - 2$ by (H8). The pool contains no term of the sequences: odd numbers, multiples of 6 and e_1 are excluded. The ends of every sequence have a least common multiple with $\rho' \geq \theta$, so each has at least k attached elements. The cycle has length $(2Q + 2) + (l - 1 - Q) + (l - Q) = 2l + 1$. $\square$

8. Proof of Theorem 1.1

Let $\varepsilon = 5 \cdot 10^{-4}$. We say that a pair (g, β) holds at n if $\text{Bad}_R(n, g) \leq (n/6)|R|\beta$ for every nonempty symmetric R . By Theorem 4.4 there is N_c such that all sixteen pairs of Table 2 hold at every $n \geq N_c$.

Lemma 8.1. *If $A \subseteq [n]$ and $|A| > T(n)$, then $|\{z \in A : z \text{ odd}\}| \geq o(n) + 1 + d + \mu_6$, $u \geq \mu_T + 1$, $d + \mu_6 \leq \mu_T$, $|U(n) \setminus A| + u + T(n) = n$ and $\mu_T \leq n/3$.*

Proof. The first bound is Lemma 2.1 with $D = (E^*(n) \cup M_6(n)) \setminus A$. The elements of A outside $U(n)$ lie in $M(n) \cap A$, which has $T(n) - \mu_T$ elements, so $u \geq T(n) + 1 - (T(n) - \mu_T)$. The third relation holds because $E^*(n)$ and $M_6(n)$ are disjoint subsets of $M(n)$, and the fourth because $|U(n)| = n - T(n)$. Finally $\mu_T + 1 \leq u \leq |U(n)|$, and $|U(n)| - 1 \leq n/3$. $\square$

Lemma 8.2. *If (g, β) holds at n and $t \leq g$, then for every $A \subseteq [n]$, $|\mathcal{O}_A(t)| \geq |\{z \in A : z \text{ odd}\}| - (n/2)\beta$, $|\mathcal{U}_A(t)| \geq u - (n/3)\beta$ and $\#\{z \in U(n) : \rho'(z) < t\} \leq (n/3)\beta$.*

Proof. Apply the definition with $R = \{1, 3, 5\}$ and $R = \{1, 5\}$; an element with $\rho' < t$ has $\rho' < g$. $\square$

Lemma 8.3 (unit boxes). *Let $|A| > T(n)$ and $2 \leq l \leq n/6$. Suppose that $\lambda_0 n \leq \mu_T \leq \lambda_1 n + 1$, that (g_T, β_T) , (g_r, β_r) , (g_t, β_t) hold at n with $\theta_T \leq g_T$, $r_1 \leq g_r$, $t_2 \leq g_t$, $r_1 \geq 0$, and that*

$$\begin{aligned} \frac{1}{2} + \frac{3}{4}\lambda_1 + \varepsilon &\leq \theta_T, & \frac{1}{6} + \frac{1}{3}\beta_T + \varepsilon &\leq \lambda_0, & \frac{1}{3}\beta_r + \varepsilon &\leq \lambda_0, \\ \frac{1}{3}\beta_t + \frac{1}{3}(1 - r_1) + \varepsilon &\leq \lambda_0, & \theta_T &\leq r_1 t_2. \end{aligned}$$

Then $G(A)$ contains a cycle of length $2l + 1$.

Table 3: The three unit boxes (Lemma 8.3).

$[\lambda_0, \lambda_1]$	θ_T	r_1	t_2
$[9/50, 23/100]$	0.6730	0.825	0.825
$[23/100, 33/100]$	0.7480	0.850	0.900
$[33/100, 1/3]$	0.7505	0.800	0.950

Proof. Let $k = \lfloor (T(n) - \mu_T + 3)/2 \rfloor$ and check the hypotheses of Proposition 7.2. Write $n = 6q + r$ with $0 \leq r < 6$; then $q \geq l$ and a direct check gives $2T(n) - n \geq 2q - 1$ and $2n - 3 \leq 3T(n) \leq 2n + 3$. By Lemma 8.1, $\mu_T + 1 \leq u \leq n - T(n)$, so $T(n) - \mu_T \geq 2T(n) - n + 1 \geq 2q$ and $k \geq q + 1 > l$. (T5) holds as $2k \geq T(n) - \mu_T + 2$. By Lemmas 8.1 and 8.2, $|\mathcal{U}_A(\theta_T)| \geq \mu_T + 1 - (n/3)\beta_T \geq (1/6 + \varepsilon)n + 1 \geq l + 1$ and $|\mathcal{U}_A(r_1)| \geq (\lambda_0 - \beta_r/3)n + 1 \geq 1$, which are (T1) and (T2). For (T3), $|U(n) \setminus A| = n - T(n) - u \leq n/3 + 1 - \mu_T - 1 \leq n/3 - \lambda_0 n$ and the bad units number at most $(n/3)\beta_r$, so the left side of (T3) is at most $2 + 2n/10^6 + n/3 - \lambda_0 n + (n/3)\beta_r \leq (n/3)r_1 + 2 + 2n/10^6 - \varepsilon n \leq (n/3)r_1$. For (T4), $k + \mu_T \leq (T(n) + 3 + \mu_T)/2 \leq n/3 + \lambda_1 n/2 + 5/2$, while $(2n/3)\theta_T \geq n/3 + \lambda_1 n/2 + (2/3)\varepsilon n$, and $(2/3)\varepsilon n \geq 5/2 + 4n/10^6$. $\square$

Lemma 8.4 (giant-path boxes). *Let $|A| > T(n)$ and $2 \leq l \leq n/6$. Suppose that $\delta_0 n \leq 3d \leq \delta_1 n$ and $a_0 n \leq \mu_6 \leq a_1 n$ with $\delta_1 \leq 9/10$; that (g_s, β_s) , (g_1, β_1) , (g_θ, β_θ) , (g_w, β_w) hold at n with $s \leq g_s$, $t_1 \leq g_1$, $\theta \leq g_\theta$, $\theta_w \leq g_w$; that $t_1, \theta_w, \kappa \geq 0$ and $s_w \leq s$; and that*

$$\begin{aligned} \frac{1}{2} + \frac{1}{2}\delta_1 + 3\kappa + \varepsilon &\leq \theta, & \frac{1}{2}\beta_s + \varepsilon &\leq \frac{1}{3}\delta_0 + a_0, & \frac{1}{2}\beta_1 + \varepsilon &\leq \frac{1}{6} + \frac{1}{3}\delta_0 + a_0, \\ \theta &\leq t_1 s, & \delta_1 + \varepsilon &\leq t_1 s_w, & \frac{1}{2}\beta_\theta + \varepsilon &\leq \frac{1}{3}\delta_0 + a_0. \end{aligned}$$

Suppose further that, unless $\kappa \geq \delta_1/6 + \varepsilon$,

$$\frac{1}{3}\beta_w + \varepsilon \leq \frac{1}{6}\delta_0 + a_0 + \kappa, \quad \frac{1}{6}\delta_1 - \kappa + a_1 + \varepsilon \leq \frac{1}{6}\theta_w s_w, \quad s_w \leq \theta_w.$$

Then $G(A)$ contains a cycle of length $2l + 1$.

Proof. Let $E = |E^*(n)|$, so $(n-3)/3 \leq E \leq (n+3)/3$, and let $k = \lfloor (E - d + 2)/2 \rfloor + \lfloor \kappa n \rfloor$. We check the hypotheses of Proposition 7.3. (H8) holds by the choice of k , and $k \geq (E - d + 1)/2 \geq 2$ since $d \leq 3n/10$. From $k \geq (E - d + 1)/2 + \kappa n - 1 \geq n/6 - d/2 + \kappa n - 1$ we get: if $\kappa \geq \delta_1/6 + \varepsilon$, then $k \geq n/6 + \varepsilon n - 1 > l$; so $k < l$ implies $\kappa < \delta_1/6 + \varepsilon$, and the second group of inequalities is available whenever $k < l$. In every case $l - k + 2 \leq d/2 - \kappa n + 3$.

(H9): $k + d \leq (E + 2 + 2\kappa n + d)/2 \leq n/6 + \delta_1 n/6 + \kappa n + 3/2$, and $(n/3)\theta \geq n/6 + \delta_1 n/6 + \kappa n + \varepsilon n/3$. (H1), (H2), (H7): by Lemmas 8.1 and 8.2, $|\mathcal{O}_A(t)| \geq o(n) + 1 + (\delta_0/3 + a_0 - \beta/2)n$ for each of the pairs used; this is at least $o(n) + 1$ for $t = s$ and $t = \theta$, which gives (H1) and (H7) as $l \leq \lfloor n/6 \rfloor \leq o(n)$, and at least $o(n) + 1 + (\varepsilon - 1/6)n \geq 3$ for $t = t_1$. (H3) is assumed. (H5): $d + 1 + n/10^6 \leq \delta_1 n/3 + 1 + n/10^6 \leq (n/3)t_1 s_w$. (H4) for $k < l$: $|\mathcal{U}_A(\theta_w)| \geq u - (n/3)\beta_w \geq d + \mu_6 + 1 - (n/3)\beta_w$, and $d + \mu_6 + 1 - (n/3)\beta_w - (d/2 - \kappa n + 3) \geq (\delta_0/6 + a_0 + \kappa - \beta_w/3)n - 2 \geq 0$. (H6) for $k < l$: $(l - k - 1) + \mu_6 \leq d/2 - \kappa n + a_1 n \leq (\delta_1/6 - \kappa + a_1)n \leq (n/6)\theta_w s_w - \varepsilon n$. $\square$

The unit boxes used are listed in Table 3 and the giant-path boxes in Table 4; Figure 9 shows the cover. In each row the thresholds g are the smallest certificate thresholds of Table 2 that are at least the corresponding parameter, and β is the certified value at g . In mode II we put $s_w = \min(\theta_w, s)$; in mode I we have $\kappa = \delta_1/6 + \varepsilon$, the last three inequalities are not needed, and we put $\theta_w = s_w = s$.

Proof of Theorem 1.1. Let $n_0 = \max(N_1, N_c)$; note $N_1 \geq N_0 \geq 13$. Let $n \geq n_0$, $A \subseteq [n]$ with $|A| > T(n)$, and $L = 2l + 1$ odd with $3 \leq L \leq n/3 + 1$, so that $1 \leq l \leq \lfloor n/6 \rfloor \leq o(n)$. If $l = 1$, apply Theorem 5.1; if $70d \leq n$, apply Theorem 6.1. Assume now $l \geq 2$ and $70d > n$, so that $\delta = 3d/n > 3/70$.

Table 4: The 29 giant-path boxes (Lemma 8.4). Decimal entries are exact except θ and κ , which are rounded to 4 decimals; the exact rationals are in the data file `boxes.json`.

$[\delta_0, \delta_1]$	$[a_0, a_1]$	mode	s	t_1	κ	θ	θ_w
[3/70, 13/140]	[0, 1/20]	II	0.650	0.850	0	0.5469	0.650
[3/70, 13/140]	[1/20, 29/175]	I	0.700	0.850	0.0160	0.5949	–
[13/140, 27/140]	[0, 1/20]	II	0.700	0.875	0.0025	0.6044	0.700
[13/140, 27/140]	[1/20, 3/40]	II	0.725	0.900	0.0150	0.6419	0.775
[13/140, 27/140]	[3/40, 313/2100]	I	0.775	0.900	0.0326	0.6949	–
[27/140, 41/140]	[0, 1/40]	II	0.750	0.875	0	0.6469	0.675
[27/140, 41/140]	[1/40, 1/20]	II	0.750	0.900	0.0075	0.6694	0.750
[27/140, 41/140]	[1/20, 3/40]	II	0.775	0.925	0.0200	0.7069	0.825
[27/140, 41/140]	[3/40, 7/80]	II	0.800	0.925	0.0250	0.7219	0.850
[27/140, 41/140]	[7/80, 3/32]	II	0.800	0.925	0.0275	0.7294	0.875
[27/140, 41/140]	[3/32, 17/160]	II	0.825	0.925	0.0325	0.7444	0.900
[27/140, 41/140]	[17/160, 7/64]	II	0.825	0.925	0.0350	0.7519	0.900
[27/140, 41/140]	[7/64, 81/700]	II	0.850	0.925	0.0400	0.7669	0.900
[41/140, 12/35]	[0, 1/40]	II	0.750	0.900	0	0.6719	0.725
[41/140, 12/35]	[1/40, 1/20]	II	0.775	0.925	0.0100	0.7019	0.775
[41/140, 12/35]	[1/20, 3/40]	II	0.800	0.925	0.0200	0.7319	0.850
[41/140, 12/35]	[3/40, 173/2100]	II	0.825	0.900	0.0200	0.7319	0.875
[12/35, 11/28]	[0, 1/40]	II	0.775	0.900	0	0.6969	0.750
[12/35, 11/28]	[1/40, 1/20]	II	0.800	0.925	0.0100	0.7269	0.800
[12/35, 11/28]	[1/20, 23/350]	II	0.825	0.900	0.0150	0.7419	0.850
[11/28, 31/70]	[0, 1/40]	II	0.800	0.925	0.0050	0.7369	0.775
[11/28, 31/70]	[1/40, 103/2100]	II	0.825	0.925	0.0125	0.7594	0.825
[31/70, 131/280]	[0, 1/80]	II	0.800	0.925	0	0.7344	0.750
[31/70, 131/280]	[1/80, 17/525]	II	0.825	0.925	0.0050	0.7494	0.800
[131/280, 29/56]	[0, 1/80]	II	0.825	0.925	0	0.7594	0.775
[131/280, 29/56]	[1/80, 3/160]	II	0.825	0.925	0	0.7594	0.800
[131/280, 29/56]	[3/160, 101/4200]	II	0.850	0.925	0.0050	0.7744	0.800
[29/56, 587/1120]	[0, 31/4200]	II	0.825	0.925	0	0.7626	0.775
[587/1120, 27/50]	[0, 89/16800]	II	0.850	0.925	0	0.7705	0.775

If $\mu_T \geq 9n/50$, then $9/50 \leq \mu_T/n \leq 1/3$ by Lemma 8.1. Split according to $\mu_T \leq \frac{23}{100}n + 1$, $\mu_T \leq \frac{33}{100}n + 1$ or neither; in each case one row of Table 3 satisfies the hypotheses of Lemma 8.3, whose inequalities are rational and are checked directly.

If $\mu_T < 9n/50$, then $d \leq \mu_T$ gives $\delta < 27/50$, and $\mu_6 \leq \mu_T - d$ gives $\mu_6/n < 9/50 - \delta/3 \leq 9/50 - \delta_0/3$ for every $\delta_0 \leq \delta$. The values of δ in $[3/70, 27/50]$ are split into the ten intervals $[\delta_0, \delta_1]$ of Table 4, and for each of them the values of μ_6/n in $[0, 9/50 - \delta_0/3]$ into the intervals $[a_0, a_1]$ listed. Every row satisfies the hypotheses of Lemma 8.4. In all cases $G(A)$ contains a cycle of length L . $\square$

9. Formal verification

The Lean development consists of 28 files and about 6600 lines; it builds with Lean 4.34.1 and Mathlib 4.34.1. The coprime graph is defined in Mathlib’s graph library: the vertices are the elements of a finite set $A \subseteq \mathbb{N}$, two vertices are adjacent when they are distinct and coprime, and a cycle of length L is a closed walk of length L without repeated vertices or edges. The statement proved is

```
def Question1 : Prop :=
  ∃ n₀ : ℕ, ∀ n ≥ n₀, ∀ A : Finset ℕ, A ⊆ Finset.Icc 1 n → T n < A.card →
    ∀ L : ℕ, Odd L → 3 ≤ L → L ≤ n / 3 + 1 → HasCycleOfLength (coprimeGraph A) L

theorem question1 : Question1
```

where $T n = n / 2 + n / 3 - n / 6$ and $/$ is division with rounding down; for an integer L the

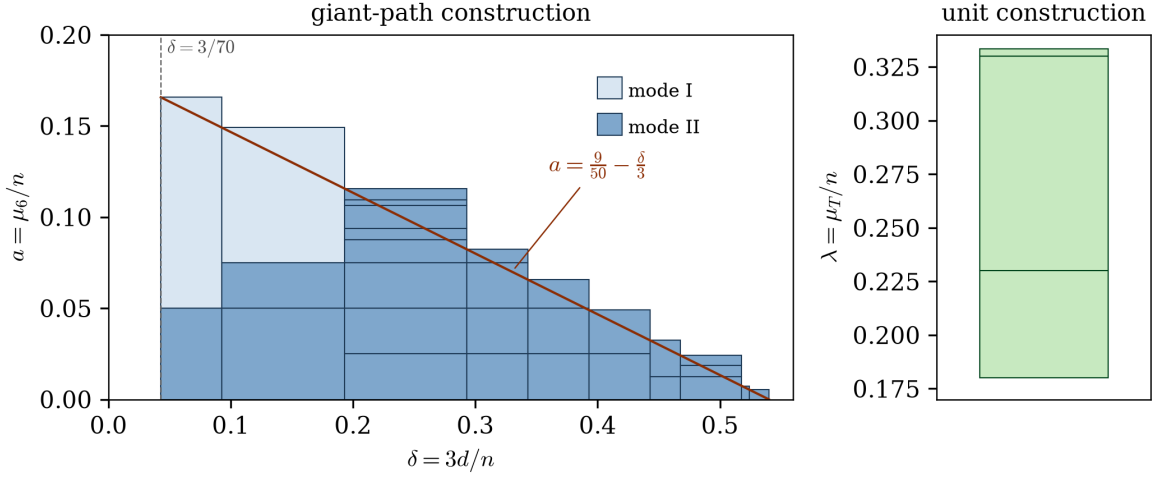


Figure 9: Left: the 29 boxes of Table 4 cover the region $3/70 \leq \delta \leq 27/50$, $0 \leq a \leq 9/50 - \delta/3$. Right: the three intervals of Table 3 cover $9/50 \leq \lambda \leq 1/3$.

condition $L \leq \lfloor n/3 \rfloor + 1$ is the same as $L \leq n/3 + 1$. The command `#print axioms question1` reports `[propext, Classical.choice, Quot.sound]`. No sorry, no additional axiom and no compiled evaluation (`native_decide`) occurs. The inequality (5), its analogue for F_{11} (54), the bounds for $\hat{F}_m$ and the sixteen certificates are evaluated in exact rational arithmetic by `decide +kernel`, and the box inequalities by the tactic `norm_num`. The file `Closure.lean`, which contains the case analysis of Section 8, is generated from the box data by a script and is checked by Lean like every other file.

The repository also contains the scripts that reproduce every number and figure of this paper: the box search and the generator of `Closure.lean` (Python), the tables, which are recomputed in exact rational arithmetic from the Lean sources (Python), and the empirical densities of Figure 4 (C).

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Conflict of interest

The authors declare no conflict of interest.

Data availability

The Lean source code, the scripts and the data supporting this article are openly available at <https://github.com/creelie/erdos-sar> and archived at Zenodo, doi:10.5281/zenodo.23151794.

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